Screen-free game kit · Design; not tested in practice

Time Detective



Learning goal

Tell today's weather apart from many years of climate information; explain why one hot or cold day does not prove a climate trend.

Materials

The card pages of this PDF (A4, 100% scale), scissors, pencils; box labels go on the table.

How to play



- No time limit; nobody is rushed.
- A wrong answer costs nothing: read why, then try again.
- A team that wants help turns over İKO's hint card.
- At the end, say what you learned in one sentence and choose the real-life task.

Task flow

Ages 7–9

Read the cards. Put the ones about today or the next days in the "WEATHER" box, and the ones about many years in the "CLIMATE" box.

1. Which box does each card go in?
2. Deniz says: "It is very hot today, so this place is always hot." What do you say?

Ages 10–12

Examine short data samples and statements. Which conclusion can really be drawn from the data?

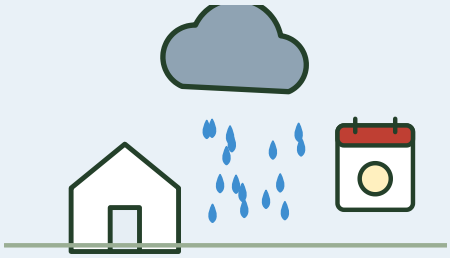
1. Measurements were taken at noon in the school garden for 7 days. Which conclusion is supported by these data?
2. Which evidence is suitable for finding out whether the climate has changed?
3. Put each statement in the right group.



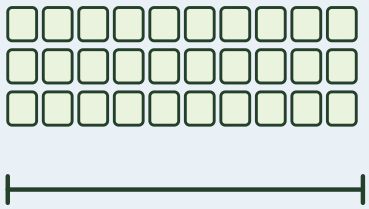
Real-life task: For one week, draw the sky at the same time each day. At the end, ask: "Does this one week show the climate?"

If a badge is used, it only shows the task was finished; it is not proof of learning.

Time Detective · Ages 7–9 · ✂ Cut along the dashed line · card 95 × 86 mm

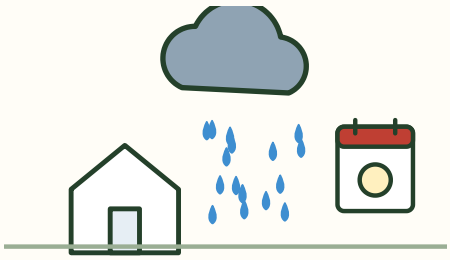


TODAY — weather



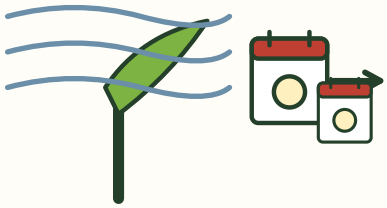
MANY YEARS — climate

Ages 7–9 · Cut-out cards



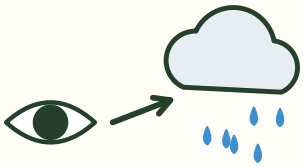
"It is raining today."

Ages 7–9 · Cut-out cards



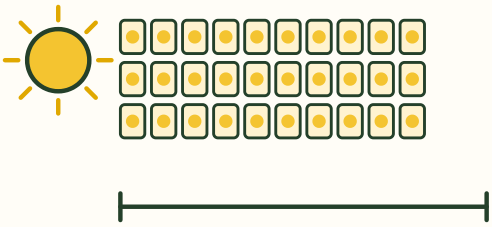
"Strong wind is expected tomorrow."

Ages 7–9 · Cut-out cards



"The sun is shining right now."

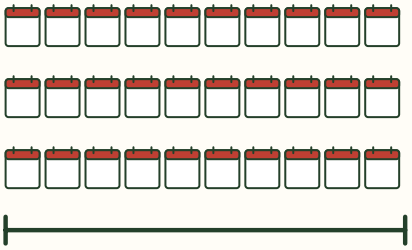
Ages 7–9 · Cut-out cards



"In this region, summers are usually dry over many years."

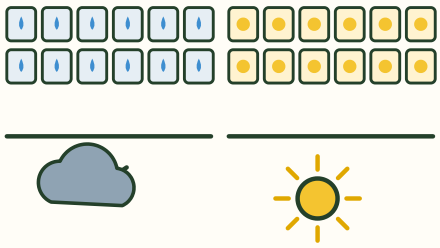
Time Detective · Ages 7–9 · ✂ Cut along the dashed line · card 95 × 86 mm

Ages 7–9 · Cut-out cards



"The July average is calculated from thirty years of records."

Ages 7–9 · Cut-out cards



"In this region, winters are usually wet over many years."

Ages 7–9 · Hint



Find the time word on the card: "today", "tomorrow", "right now", or "many years", "thirty years"?

Ages 7–9 · Question card

Deniz says: "It is very hot today, so this place is always hot." What do you say?

A. One day does not show the climate; we need records over many years.

B. Right, if today is hot, the climate is hot.

Ages 7–9 · Hint



One day, or many years? Which do we need to understand climate?

Time Detective · Ages 10–12 · ✂ Cut along the dashed line · card 95 × 86 mm

Ages 10–12 · Question card

Measurements were taken at noon in the school garden for 7 days. Which conclusion is supported by these data?

SAMPLE DATA — not a measurement from a real place

Day	1	2	3	4	5	6	7
Noon, °C	31	33	30	34	32	35	33

A. This week, the noon values measured in this garden were 30 °C or more.

B. This city's climate is warming.

C. Next summer will be the same.

Ages 10–12 · Hint



Where and for how long were the data collected? Does the conclusion go beyond that place and time?

Ages 10–12 · Question card

Which evidence is suitable for finding out whether the climate has changed?

SAMPLE DATA — not a measurement from a real place

Period	July average
Period A (30 years)	22.5 °C
Period B (next 30 years)	23.4 °C

A. Comparing the averages of two 30-year periods measured the same way.

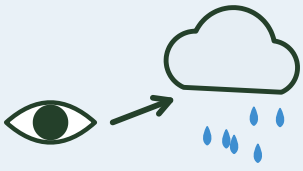
B. Finding the single hottest day.

C. Looking at last week's average.

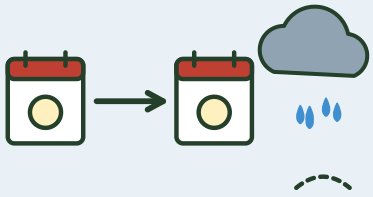
Ages 10–12 · Hint



How many years are averaged for climate?

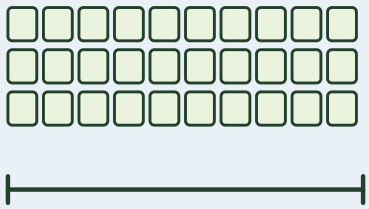


Weather observation

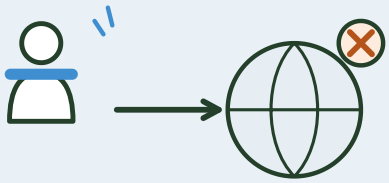


Weather forecast

Time Detective · Ages 10–12 · ✂ Cut along the dashed line · card 95 × 86 mm



Climate



Faulty inference

Ages 10–12 · Cut-out cards

"Last week we took a measurement in the garden every day at noon."

Ages 10–12 · Cut-out cards

"Strong wind is expected tomorrow."

Ages 10–12 · Cut-out cards

"According to thirty years of records, winters in this city are wetter than summers."

Ages 10–12 · Cut-out cards

"According to the IPCC, global surface temperature in 2011–2020 was about 1.1 °C higher than in 1850–1900." (K01)

Time Detective · Ages 10–12 · ✂ Cut along the dashed line · card 95 × 86 mm

Ages 10–12 · Cut-out cards

"Yesterday was very cold, so there is no global warming."

Ages 10–12 · Cut-out cards

"This week was very hot, so the climate has definitely changed."

Ages 10–12 · Hint



Find the time scale first. Then: observation, forecast, long-term pattern, or a big conclusion from a short observation?

Answers and explanations (teacher only)

Ages 7–9

1	“It is raining today.”	TODAY — weather	“Today” is a short time: this is weather.
1	“Strong wind is expected tomorrow.”	TODAY — weather	It is a weather forecast for “tomorrow”.
1	“The sun is shining right now.”	TODAY — weather	“Right now” is observed weather.
1	“In this region, summers are usually dry over many years.”	MANY YEARS — climate	“Over many years” describes a pattern: this is climate.
1	“The July average is calculated from thirty years of records.”	MANY YEARS — climate	A thirty-year average is the basis of a climate normal.
1	“In this region, winters are usually wet over many years.”	MANY YEARS — climate	Again a pattern over many years: climate.
2A	One day does not show the climate; we need records over many years.	Fits	Yes. Climate is the average state over many years.
2B	Right, if today is hot, the climate is hot.	Does not fit	Today only shows the weather. Tomorrow may be cool. Climate needs many years.

Ages 10–12

1A	This week, the noon values measured in this garden were 30 °C or more.	Fits	The conclusion stays within the place and time of the data.
1B	This city’s climate is warming.	Does not fit	Seven days at one spot are not enough for a climate trend. Long-term records measured the same way are needed.
1C	Next summer will be the same.	Does not fit	These data say nothing about the future.
2A	Comparing the averages of two 30-year periods measured the same way.	Fits	Yes. Climate normals are based on 30-year periods (K07). For a real conclusion, the official source, station and period must be stated.
2B	Finding the single hottest day.	Does not fit	One day is weather; it does not show long-term change.
2C	Looking at last week’s average.	Does not fit	A week is also a short time.
3	“Last week we took a measurement in the garden every day at noon.”	Weather observation	Measurements over a short time: weather observation.
3	“Strong wind is expected tomorrow.”	Weather forecast	An expectation for the next day: forecast.
3	“According to thirty years of records, winters in this city are wetter than summers.”	Climate	A pattern over many years: climate.

3	“According to the IPCC, global surface temperature in 2011–2020 was about 1.1 °C higher than in 1850–1900.” (K01)	Climate	A global, long-term comparison: climate. Source: IPCC AR6 Synthesis Report (K01).
3	“Yesterday was very cold, so there is no global warming.”	Faulty inference	A global, long-term conclusion cannot be drawn from one day in one place.
3	“This week was very hot, so the climate has definitely changed.”	Faulty inference	One hot week alone does not prove a climate trend.

Time Detective: Weather is short-term: today, tomorrow, this week. Climate is the average state over many years, usually 30. One day does not show the climate.